



# Screen Machine: Pre-Snap Play Prediction in the NFL

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## Background

**Motivation:**  
Since 2010, the NFL has witnessed an **upward trend** in **screen passes**. Due to their quick execution and reliable nature, being able to predict screen passes is a valuable defensive edge.

- Goals:**  
Predict the likelihood of a screen pass **before** the snap
- Identify lateral direction (left or right)
  - Identify target receiver group (WR or non-WR)

**Defining a Screen Pass:**  
A quick forward or lateral pass that utilizes surrounding blockers to protect the ball carrier

- Data Overview:**  
Modeling Dataset (NFL Big Data Bowl 2025):
- 2022 Regular Season Weeks 1-9
  - Game, play-by-play, player-play, tracking
- Supplementary Dataset (nflfastR, nflreadr):
- Historical play-by-play 2020-22

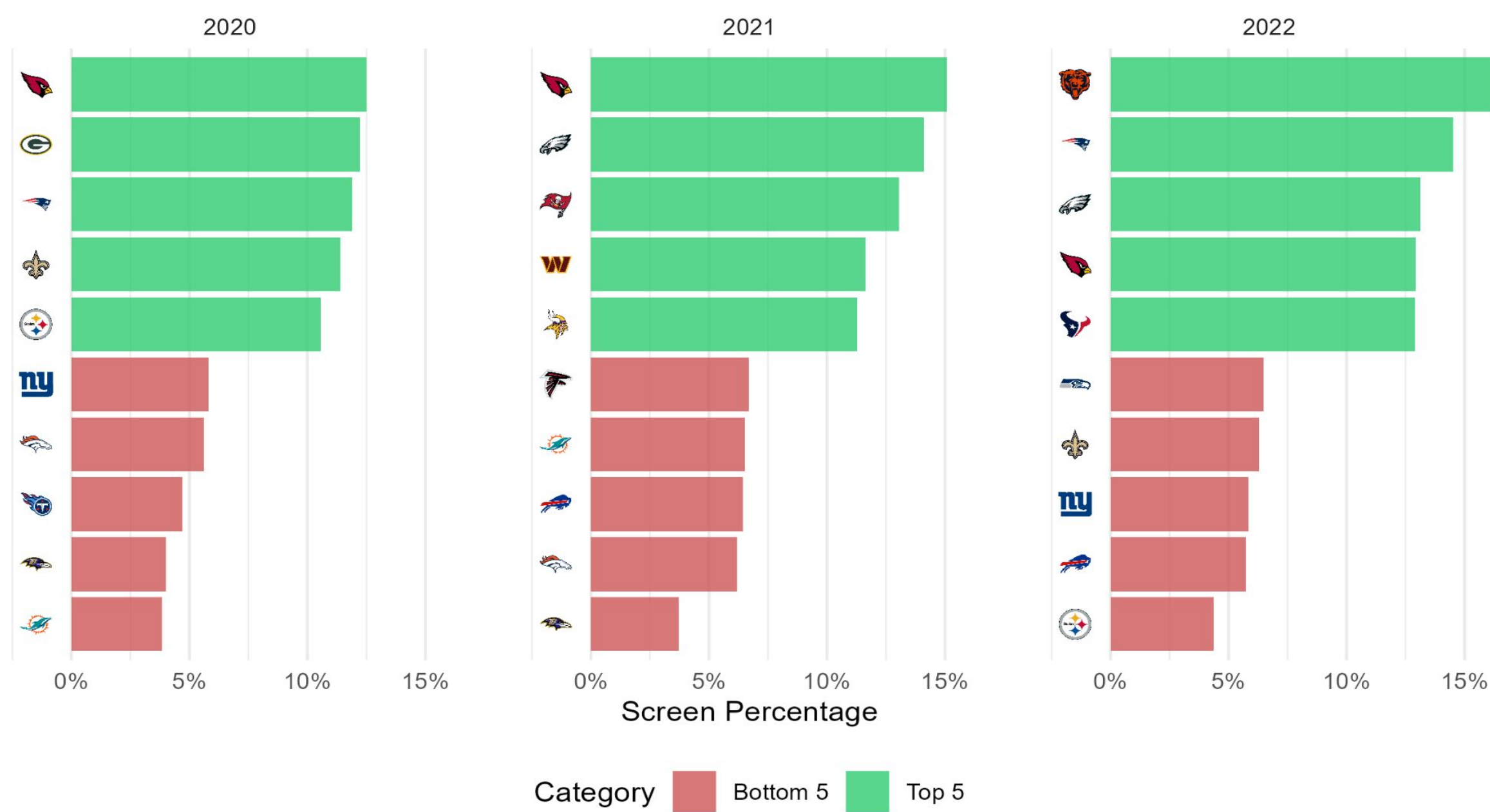
## Exploratory Data Analysis

There is No Significant Difference Between Share of Screen Targets by Position & Direction

|       | WR          | RB/TE       | Total        |
|-------|-------------|-------------|--------------|
| Left  | 196 (48.9%) | 205 (51.1%) | 401 (53.5%)  |
| Right | 182 (52.3%) | 166 (47.7%) | 348 (46.5%)  |
| Total | 378 (50.5%) | 371 (49.5%) | 749 (100.0%) |

Chi-square test of independence:  $X^2 = 0.74$ ,  $df = 1$ ,  $p = 0.389$

### Team Screen Tendencies Across Seasons



## Methodology

### Modeling Approach:

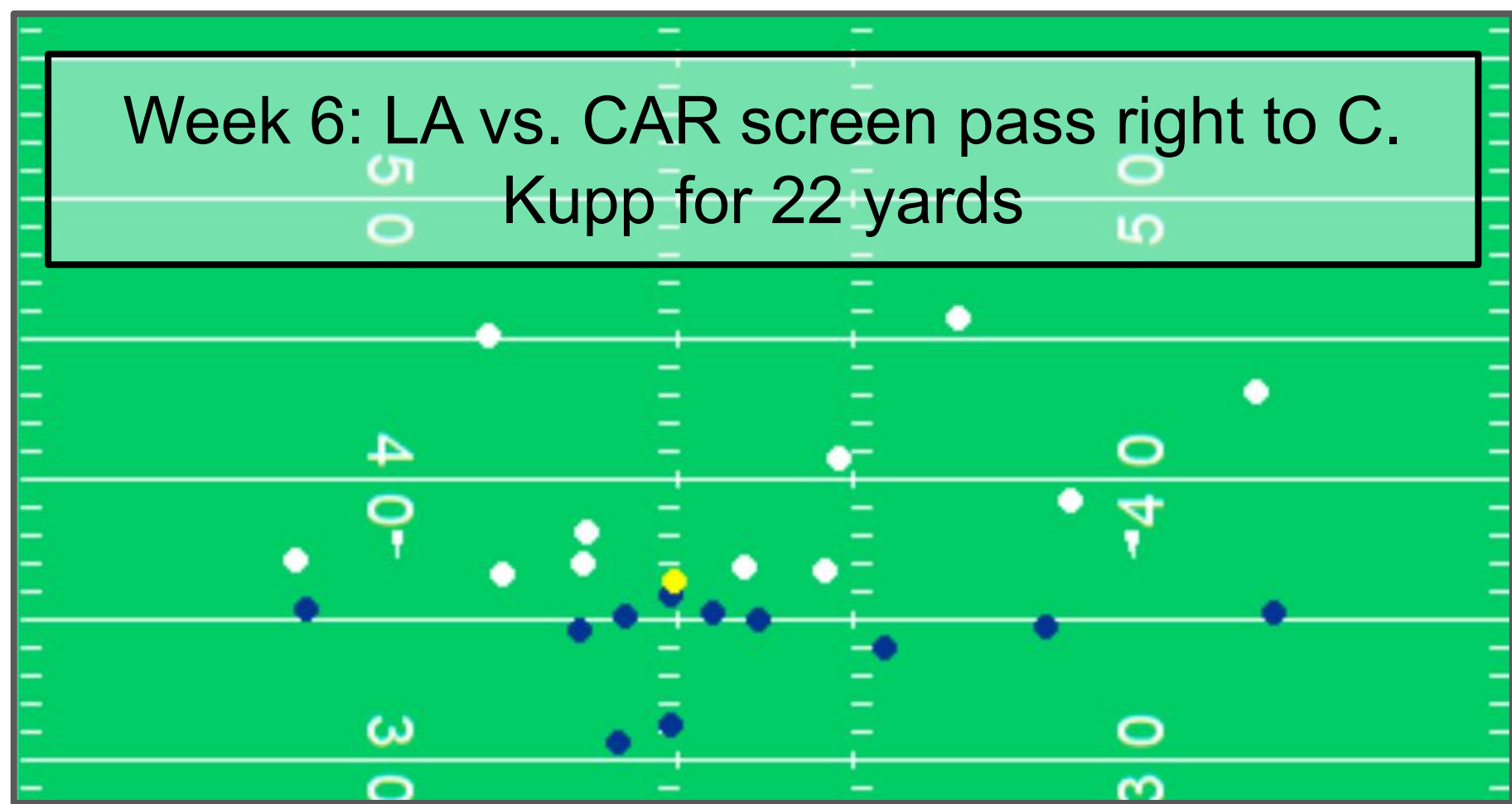
Screen Prediction Model

Lateral Direction Model

Receiver Group Model

### Feature Set Overview:

- Screen Prediction:
- Play context
  - Historical team tendencies (based on 2020-21)
- Direction & Receiver Group:
- Receiver split
  - Spatial features (e.g., WR distance to QB)



### Model Evaluation: (Baseline: Naive Guess Rate)

| Screen Model | Accuracy     | Kappa        | ROC AUC      |
|--------------|--------------|--------------|--------------|
| XGBoost      | 0.675 ± 0.01 | 0.268 ± 0.02 | 0.749 ± 0.02 |
| LASSO        | 0.589 ± 0.01 | 0.129 ± 0.01 | 0.684 ± 0.01 |

12.1% Baseline

51.6% Baseline

| Lateral Model | Accuracy     | Kappa        | ROC AUC      |
|---------------|--------------|--------------|--------------|
| XGBoost       | 0.703 ± 0.03 | 0.406 ± 0.05 | 0.758 ± 0.03 |
| LASSO         | 0.684 ± 0.02 | 0.364 ± 0.03 | 0.747 ± 0.03 |

50.3% Baseline

| Receiver Model | Accuracy     | Kappa        | ROC AUC      |
|----------------|--------------|--------------|--------------|
| XGBoost        | 0.681 ± 0.03 | 0.366 ± 0.06 | 0.748 ± 0.04 |
| LASSO          | 0.593 ± 0.02 | 0.184 ± 0.04 | 0.630 ± 0.02 |

## Results

### Single-Play Model Application



Possession Team: Rams | Context: 3rd & 13

| Feature                   | Value / Status |
|---------------------------|----------------|
| <b>Screen Probability</b> |                |
| Screen                    | 58.9%          |
| Not Screen                | 41.1%          |
| <b>Direction</b>          |                |
| Left                      | 37.1%          |
| Right                     | 62.9%          |
| <b>Receiver Group</b>     |                |
| WR                        | (55.8%)        |
| Non-WR                    | (44.2%)        |

### Lateral Direction Predictability by Team

| Team            | Accuracy - baseline | Kappa | n  |
|-----------------|---------------------|-------|----|
| <b>Top 5</b>    |                     |       |    |
| PHI             | 32.4%               | 0.776 | 37 |
| TB              | 31.4%               | 0.658 | 35 |
| SF              | 29.0%               | 0.612 | 31 |
| NYJ             | 24.0%               | 0.590 | 25 |
| WAS             | 22.9%               | 0.535 | 35 |
| <b>Bottom 5</b> |                     |       |    |
| GB              | 6.9%                | 0.171 | 29 |
| HOU             | 4.0%                | 0.237 | 25 |
| CAR             | 3.4%                | 0.151 | 29 |
| CHI             | -2.9%               | 0.235 | 34 |
| ATL             | -4.2%               | 0.229 | 24 |

### Receiver Group Predictability by Team

| Team            | Accuracy - baseline | Kappa | n  |
|-----------------|---------------------|-------|----|
| <b>Top 5</b>    |                     |       |    |
| NE              | 26.5%               | 0.529 | 34 |
| ATL             | 20.8%               | 0.417 | 24 |
| ARI             | 20.0%               | 0.458 | 45 |
| HOU             | 16.0%               | 0.426 | 25 |
| CAR             | 13.8%               | 0.394 | 29 |
| <b>Bottom 5</b> |                     |       |    |
| CHI             | 8.8%                | 0.304 | 34 |
| PHI             | 2.7%                | 0.198 | 37 |
| GB              | 0.0%                | 0.340 | 28 |
| TB              | -2.9%               | 0.330 | 35 |
| MIN             | -13.0%              | 0.115 | 23 |

## Discussion

### Conclusion:

- Developed an XGBoost/LASSO modeling framework to predict screen pass **decisions** using **pre-snap** features
- Identified the Eagles and Buccaneers as the most predictable screen pass teams by spatial features

### Limitations:

- Sparse number of single-season screen plays per team
- Player turnovers can change team tendencies

### Future Work:

- Use sequential models for real-time screen probability
- Find individual teams' pre-snap "tells" with tracking data

## Acknowledgements

Special thanks to Dr. Karim Kassam, Erin Franke, Sara Colando, Dr. Ron Yurko, Quang Nguyen, and CMSACamp TAs for their guidance throughout the research process.